Biology

Chapter 3: Movement into and out of the Cell

Welcome to AnithUncommon's **Sample Resources!**

In Biology , 3.1…

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by Yassmin Younes to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in Biology O'Level join our nonprofit!

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This sample is made thanks to, Yassmin Younes

— Made by students, for students.

Lesson Objectives:

At the end of the lesson, students should be *able* to…

· Explain the net movement of particles down a concentration gradient and identify the specific factors.

· Predict the direction of water movement between solutions of different water potentials and explain the structural consequences for cells

· Contrast the "free" movement of molecules with the energy-demanding process of moving substances against a concentration gradient using carrier proteins and energy.

· Understanding how the "Surface Area to Volume Ratio" affects the rate of movement into and out of specialized cells.

Reflective Questions to Consider:

Molecules are in a state of constant, chaotic motion, yet this randomness is the very thing that allows life to sustain itself. For an organism to survive, it must navigate the "traffic control" of the cell membrane, deciding which substances can drift in for free and which require a physical "pump" to move.

Below are the following core questions you will face as we explore this chapter:

- Since Diffusion and Osmosis rely entirely on the "free" kinetic energy of moving particles, how does a cell maintain its internal balance when the environment outside is constantly trying to change it?
- If water always moves toward the lowest water potential, to what extent is the survival of a plant dependent on its ability to create "tension" between its roots and the surrounding soil?
- Active Transport allows a cell to move molecules "uphill" against the natural flow of physics—why would a cell choose to spend its precious metabolic energy on this difficult process rather than waiting for things to arrive naturally?

Syllabus Outline:

Unit 3: Movement into and out of the Cell
Tip! • **You may find it difficult to keep the directions of movement straight, especially when comparing "down" a gradient versus "against" one. The secret is to link the movement to the cell's energy budget: if it's "free" (Diffusion/Osmosis), the molecules are just following the crowd; if it's "expensive" (Active Transport), the cell is working hard to grab what it needs.**

Activities in Class:

- **Lesson Breakdown**
- **Create a diagrams to visualize the random movement of particles in gases and liquids, illustrating the moment a system reaches equilibrium and net movement stops.**
- **Hand-draw the physical transformation of an animal cell and a plant cell when placed in distilled water versus a concentrated salt solution.**

3.1 Diffusion	Mentor's Notes:
The Fundamentals **a. Definition** → The **net movement** of particles from a region of their **higher concentration** to a region of their **lower concentration** down a **concentration gradient**, as a result of their **random movement**. **b. Energy Requirement** → Diffusion is a **passive process**. It does not require energy from the cell (ATP) because it uses the kinetic energy already present in the moving particles. **Diffusion and the Cell Membrane** **a. The Gateway** → For a substance to enter or leave a cell, it must pass through the **cell surface membrane**. → Only small molecules (like oxygen, carbon dioxide, and water) can diffuse directly through the membrane. Large molecules (like starch or proteins) are too "bulky" and cannot pass through. **b. Practical Examples in Life** • **Gas Exchange:** Oxygen diffuses from the air spaces in the lungs (high concentration) into the blood (low concentration). • **Excretion:** Carbon dioxide diffuses from the cells (high concentration) into the blood to be carried away.	**Try to RECOGNIZE the importance of the Concentration Gradient.** **The steeper the "hill" (the bigger the difference in concentration between two areas), the faster the molecules will "slide" down it.** **In exams, if a question asks how to speed up diffusion, always think about making that gradient steeper!**

Factors that Affect the Rate of Diffusion

Factor	Change	Effect on Diffusion Rate	Why?
Surface Area	Increase	**Faster**	More "doors" for the particles to pass through at once.
Temperature	Increase	**Faster**	Particles gain more kinetic energy and move more quickly.
Concentration Gradient	Steeper	**Faster**	The "push" from the crowded area is much stronger.
Distance	Decrease	**Faster**	Particles have a shorter "trip" to take to reach their destination.

When defining diffusion, never forget the phrase "Net Movement." Molecules are always moving in both directions, but "net movement" refers to the overall direction that most of them are going. Including this specific term shows the examiner you have a high-level understanding of molecular behavior.

Direction: From High to Low (Down the gradient).

Medium: Can happen in gases and liquids (where particles are free to move).

Driving Force: Random kinetic energy of particles.

Membrane: Must be permeable to the specific molecule.

3.2 Osmosis **The Definition** **a. Scientific Definition** → The **net movement** of **water molecules** from a region of **higher water potential** (dilute solution) to a region of **lower water potential** (concentrated solution), through a **partially permeable membrane**. **b. The Partially Permeable Membrane** → Think of the cell membrane as a "sieve" with tiny holes. Water molecules are small enough to pass through, but large solute molecules (like glucose or salt) are too big and get left behind. **Water Potential (Ψ)** • **High Water Potential:** Pure water or a very dilute solution (lots of "free" water molecules). • **Low Water Potential:** A concentrated solution (water molecules are "busy" surrounding solute particles). • **Movement:** Water always moves **down** a water potential gradient (from High Ψ to Low Ψ).	**Try to RECOGNIZE that Osmosis is just Diffusion with a specific name.** **If the question mentions "water" and a "partially permeable membrane," the answer is Osmosis.** **If it mentions "oxygen" or "perfume," it is Diffusion.** **Using the correct term is a must for scoring points in the "Short Answer" sections!** **Avoid saying "concentration of water" in your exam. Instead, use the term "Water Potential."** **Ψ this represents water potential.**

Osmosis in Animal Cells (The Danger)

Solution Type	Environment vs. Cell	Result for Animal Cell
Hypotonic	Higher Water Potential outside	Water enters, Cell swells and **bursts** (Lysis).
Isotonic	Equal Water Potential	No net movement, Cell stays **normal**.
Hypertonic	Lower Water Potential outside	Water leaves, Cell **shrivels** (Crenation).

Osmosis in Plant Cells (The Support)

Plant cells have a strong **Cellulose Cell Wall**, which prevents them from bursting and provides support.

a. Turgid (The Happy State)

→ When a plant cell is in pure water, water enters by osmosis. The vacuole swells and pushes against the cell wall. The cell becomes **Turgid**. This pressure (Turgor Pressure) keeps the plant upright.

Animal cells lack a cell wall, which makes them very sensitive to water movement.

A solution with a lot of water has a High Water Potential, and a salty/sugary solution has a Low Water Potential. Cambridge examiners love this terminology because it shows a higher level of scientific maturity.

b. Flaccid & Plasmolysed (The Wilting State)

→ If a plant loses water, the vacuole shrinks. The cell becomes **Flaccid** (soft).

→ If it loses too much water, the cell membrane pulls away from the cell wall. This is called **Plasmolysis**.

Comparison: Animal vs. Plant Osmosis

Feature	**Animal Cell**	**Plant Cell**
Outer Boundary	Flexible Membrane only	Rigid Cell Wall
Reaction to Pure Water	Bursts (Lysis)	Becomes Turgid (Strong)
Reaction to Salt Water	Shrivels	Plasmolyses
Main Organelle Involved	Cytoplasm	Large Permanent Vacuole

3.3 Active Transport

The Definition

a. Scientific Definition

→ The movement of particles through a cell membrane from a region of **lower concentration** to a region of **higher concentration** (i.e., **against** a concentration gradient), using **energy from respiration**.

Try to RECOGNIZE the role of Mitochondria in this process.

b. The Mechanism

→ This process requires specialized **Carrier Proteins** embedded in the cell membrane. These proteins act like "pumps" that pick up specific molecules and force them through to the other side.

Since Active Transport requires energy, any cell doing a lot of it (like a Root Hair cell or an Epithelial cell in the gut) will be packed with mitochondria.

Why is Energy Needed?

Unlike Diffusion and Osmosis, which are "passive" and happen for free, Active Transport is "active."

- **Source of Energy:** Respiration (happening in the mitochondria).
- **The "Work":** The energy is used to change the shape of the carrier protein to move the molecule across the membrane.

If an exam question shows a cell with many "powerhouses" and asks how it takes in minerals, the answer is Active Transport!

Real-Life Examples (The "Necessities")

a. Root Hair Cells (Plants)

→ The concentration of mineral ions (like nitrates) is often higher inside the root hair cell than in the surrounding soil water. The plant uses Active Transport to "pump" these ions into the root so it can grow.

b. Small Intestine (Humans)

→ After a meal, glucose is absorbed into the blood. Even when the concentration in the blood becomes higher than in the gut, the body continues to absorb every last molecule of glucose using Active Transport.

Always use the phrase "Against a

Comparison: Passive vs. Active Transport

Feature	Diffusion	Osmosis	Active Transport
Direction	High to Low	High Ψ to Low Ψ	Low to High
Gradient	**Down**	**Down**	**Against**
Energy (ATP)	❌ Not required	❌ Not required	✔ Required
Protein Pump	❌ Not required	❌ Not required	✔ Required
Substance	Gases / Solutes	**Water only**	Ions / Nutrients

concentration gradient" in your descriptions. This is the key technical marker Cambridge looks for.

Mentors' Sources:

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